

Chapter 2

Chi-square Distribution

2.1 χ^2 -variate

The square of a standard normal variate is known as a chi-square variate with 1 degree of freedom (*d.f.*).

Thus if $X \sim N(\mu, \sigma^2)$, then $Z = \frac{X-\mu}{\sigma} \sim N(0, 1)$ and

$$Z^2 = \left(\frac{X - \mu}{\sigma} \right)^2,$$

is a χ^2 -variate with 1 d.f.

In general if $X_i, (i = 1, 2, \dots, n)$ are n independent normal variates with means μ_i and variances $\sigma_i^2, (i = 1, 2, \dots, n)$, then

$$\chi^2 = \sum_{i=1}^n \left(\frac{X_i - \mu_i}{\sigma_i} \right)^2,$$

is a χ^2 -variate with n d.f.

The χ^2 -distribution with n degrees of freedom is the distribution of a sum of the squares of n independent standard normal random variables. This distribution is a special case of the gamma distribution and is one of the most widely used probability distributions in inferential statistics, notably in hypothesis testing and in construction of confidence intervals.

2.2 Derivation of χ^2 -distribution

Method of Moment Generating Function (MGF)

If $X_i, (i = 1, 2, \dots, n)$ are independent $N(\mu_i, \sigma_i^2)$, we want the distribution of

$$\chi^2 = \sum_{i=1}^n \left(\frac{X_i - \mu_i}{\sigma_i} \right)^2 = \sum_{i=1}^n U_i^2, \quad \text{where } U_i = \frac{X_i - \mu_i}{\sigma_i} \sim N(0, 1).$$

Since X_i 's are independent, so we have

$$\begin{aligned} M_{\chi^2}(t) &= M_{\sum_{i=1}^n U_i^2}(t) \\ &= \prod_{i=1}^n M_{U_i^2}(t) \\ &= \left[M_{U_i^2}(t) \right]^n. \end{aligned} \tag{2.1}$$

Hence, U_i 's are independent and identically distributed (*i.i.d.*) standard normal variate $N(0, 1)$. Now we can obtain as

$$\begin{aligned} M_{U_i^2}(t) &= E \left[e^{tU_i^2} \right] = \int_{-\infty}^{\infty} e^{tu_i^2} f(x_i) dx_i \\ &= \int_{-\infty}^{\infty} e^{tu_i^2} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x_i-\mu}{\sigma}\right)^2} dx_i \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tu_i^2} e^{-\frac{u_i^2}{2}} du_i \quad \left[u_i = \frac{x_i - \mu}{\sigma} \Rightarrow dx_i = \sigma du_i \right] \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\left(\frac{1-2t}{2}\right)u_i^2} du_i \\ &= \frac{1}{\sqrt{2\pi}} \frac{\sqrt{2\pi}}{\sqrt{1-2t}} \left\{ \int_{-\infty}^{\infty} \frac{1}{\frac{1}{\sqrt{1-2t}} \cdot \sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{u_i-0}{\frac{1}{\sqrt{1-2t}}}\right)^2} du_i \right\} \\ &= (1-2t)^{-\frac{1}{2}}. \end{aligned}$$

By using (2.1), we have

$$M_{\chi^2}(t) = (1-2t)^{-\frac{n}{2}} = \left(1 - \frac{t}{\frac{1}{2}} \right)^{-\frac{n}{2}},$$

which is moment generating function (*m.g.f.*) of the Gamma variate with model

parameters $\frac{n}{2}$ and $\frac{1}{2}$. Hence by uniqueness theorem of *m.g.f.*'s,

$$\chi^2 = \sum_{i=1}^n \left(\frac{X_i - \mu_i}{\sigma_i} \right)^2,$$

is a Gamma variate with model parameters $\frac{n}{2}$ and $\frac{1}{2}$, and further expressed as

$$\begin{aligned} f(\chi^2) &= \frac{\left(\frac{1}{2}\right)^{\frac{n}{2}} e^{-\frac{1}{2}\chi^2} (\chi^2)^{\frac{n}{2}-1}}{\Gamma\left(\frac{n}{2}\right)} \\ &= \frac{1}{2^{\frac{n}{2}} \Gamma\left(\frac{n}{2}\right)} e^{-\frac{\chi^2}{2}} (\chi^2)^{\frac{n}{2}-1}, \quad 0 \leq \chi^2 \leq \infty, \end{aligned}$$

which is the required density function of χ^2 -distribution with n degrees of freedom.

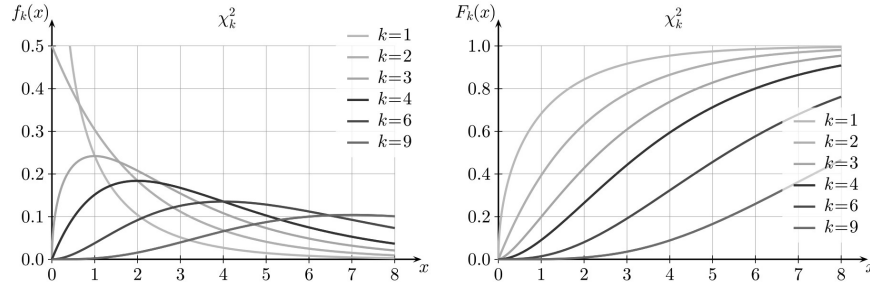


Figure 2.1: The pdf and cdf of the χ^2 -distribution.

Theorem: If $X \sim \chi_{(n)}^2$, then $\frac{X}{2} \sim \gamma\left(\frac{n}{2}\right)$.

Proof: Here given as $X \sim \chi_{(n)}^2$, so

$$f(x) = \frac{1}{2^{\frac{n}{2}} \Gamma\left(\frac{n}{2}\right)} e^{-\frac{x}{2}} x^{\frac{n}{2}-1}. \quad (2.2)$$

Let $Y = \frac{X}{2} \Rightarrow \frac{dx}{dy} = 2$. By using (2.2), we can write as

$$\begin{aligned} f(y) &= \frac{1}{2^{\frac{n}{2}} \Gamma\left(\frac{n}{2}\right)} e^{-y} (2y)^{\frac{n}{2}-1} \cdot 2 \\ &= \frac{e^{-y} y^{\frac{n}{2}-1}}{\Gamma\left(\frac{n}{2}\right)}; \quad 0 \leq y \leq \infty. \end{aligned}$$

$\therefore Y = \frac{X}{2} \sim \gamma\left(\frac{n}{2}\right)$.

Remarks:

- (i) If $X \sim \chi^2(2)$, then $X \sim \text{Exp}(1/2)$ is an exponential distribution.
- (ii) If $X \sim \text{Rayleigh}(1)$, then $X^2 \sim \chi^2(2)$.
- (iii) If $X \sim \chi^2(n_1)$ and $Y \sim \chi^2(n_2)$ are independent then $\frac{X}{X+Y} \sim \text{Beta}(\frac{n_1}{2}, \frac{n_2}{2})$.
- (iv) If $X \sim U(0, 1)$, then $-2\log(X) \sim \chi^2(2)$.

and so on.

2.3 Moment Generating Function of χ^2 -distribution

Let $X \sim \chi^2_{(n)}$, then

$$\begin{aligned}
 M_X(t) &= E[e^{tX}] \\
 &= \int_0^\infty e^{tx} f(x) dx \\
 &= \int_0^\infty e^{tx} \frac{1}{2^{\frac{n}{2}} \Gamma(\frac{n}{2})} e^{-\frac{x}{2}} (x)^{\frac{n}{2}-1} dx \\
 &= \frac{1}{2^{\frac{n}{2}} \Gamma(\frac{n}{2})} \int_0^\infty e^{-(\frac{1-2t}{2})x} (x)^{\frac{n}{2}-1} dx \\
 &= \frac{1}{2^{\frac{n}{2}} \Gamma(\frac{n}{2})} \frac{\Gamma(\frac{n}{2})}{(\frac{1-2t}{2})^{\frac{n}{2}}} \quad [\text{Using Gamma Integral}] \\
 &= (1-2t)^{-\frac{n}{2}}, \quad |2t| < 1,
 \end{aligned}$$

which is the moment generating function of a χ^2 -variate with n degrees freedom.

2.4 Cumulant Generating Function of χ^2 -distribution

Let $X \sim \chi^2_{(n)}$, then

$$\begin{aligned}
 K_X(t) &= \log \{M_X(t)\} \\
 &= -\frac{n}{2} \log(1-2t)
 \end{aligned}$$

$$\begin{aligned}
&= -\frac{n}{2} \left[-2t - \frac{(2t)^2}{2} - \frac{(2t)^3}{3} - \frac{(2t)^4}{4} - \dots - \frac{(2t)^r}{r} - \dots \right] \\
&= nt + 2n \frac{t^2}{2!} + 8n \frac{t^3}{3!} + 48n \frac{t^4}{4!} + \dots + n2^{r-1}(r-1)! \frac{t^r}{r!} + \dots \quad (2.3)
\end{aligned}$$

In general, k_r = Coefficient of $\frac{t^r}{r!}$ in $K_X(t) = n2^{r-1}(r-1)!$. Therefore,

$$\begin{aligned}
\text{Mean} &= k_1 = n, \\
\mu_2 &= k_2 = 2n, \\
\mu_3 &= k_3 = 8n, \\
\mu_4 &= k_4 + 3k_2^2 = 48n + 12n^2, \\
\beta_1 &= \frac{\mu_3^2}{\mu_2^3} = \frac{64n^2}{8n^3} = \frac{8}{n}, \text{ and} \\
\beta_2 &= \frac{\mu_4}{\mu_2^2} = \frac{48n + 12n^2}{4n^2} = \frac{12}{n} + 3.
\end{aligned}$$

2.5 Characteristic Function of χ^2 -distribution

Let $X \sim \chi_{(n)}^2$, then

$$\begin{aligned}
M_X(t) &= E[e^{itX}] \\
&= \int_0^\infty e^{itx} f(x) dx \\
&= \int_0^\infty e^{itx} \frac{1}{2^{\frac{n}{2}} \Gamma\left(\frac{n}{2}\right)} e^{-\frac{x}{2}} (x)^{\frac{n}{2}-1} dx \\
&= \frac{1}{2^{\frac{n}{2}} \Gamma\left(\frac{n}{2}\right)} \int_0^\infty e^{-\left(\frac{1-2it}{2}\right)x} (x)^{\frac{n}{2}-1} dx \\
&= \frac{1}{2^{\frac{n}{2}} \Gamma\left(\frac{n}{2}\right)} \frac{\Gamma\left(\frac{n}{2}\right)}{\left(\frac{1-2it}{2}\right)^{\frac{n}{2}}} \quad [\text{Using Gamma Integral}] \\
&= (1-2it)^{-\frac{n}{2}}, \quad |2it| < 1,
\end{aligned}$$

which is the characteristic function of a χ^2 -variate with n degrees freedom.

2.6 Mode and Skewness of χ^2 -distribution

Let $X \sim \chi^2_{(n)}$, then

$$f(x) = \frac{1}{2^{\frac{n}{2}} \Gamma\left(\frac{n}{2}\right)} e^{-\frac{x}{2}} (x)^{\frac{n}{2}-1}, \quad 0 \leq x < \infty. \quad (2.4)$$

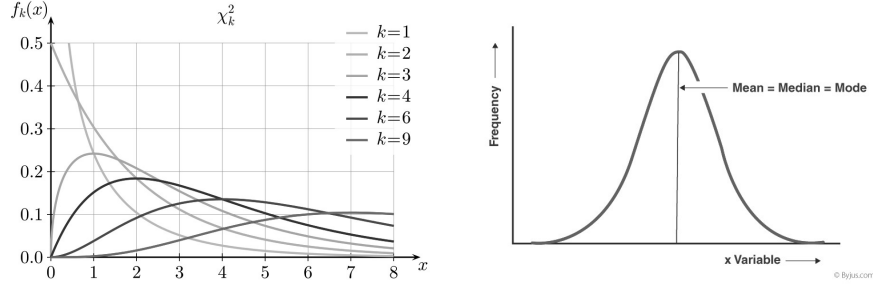


Figure 2.2: The pdf of normal distribution.

Mode of the distribution is the solution of $f'(x) = 0$ and $f''(x) < 0$.

Logarithmic differentiation with respect to x in (2.4) gives:

$$\begin{aligned} \log(f(x)) &= \log(c) - \frac{x}{2} + \left(\frac{n}{2} - 1\right) \log(x) \quad \left[c = \frac{1}{2^{\frac{n}{2}} \Gamma\left(\frac{n}{2}\right)} \right] \\ \therefore \frac{f'(x)}{f(x)} &= 0 - \frac{1}{2} + \left(\frac{n}{2} - 1\right) \cdot \frac{1}{x} = \frac{n-2-x}{2x}. \end{aligned}$$

Since $f(x) \neq 0$,

$$\begin{aligned} f'(x) &= 0 \\ \Rightarrow \frac{n-2-x}{2x} &= 0 \\ \therefore x &= n-2. \end{aligned}$$

It can be easily seen that at the point, $x = n-2$, $f''(x) < 0$. Hence the mode of the χ^2 -distribution with n degrees freedom is $(n-2)$.

Also Karl Pearson's coefficient of skewness is given by:

$$\begin{aligned}\text{Skewness} &= \frac{\text{Mean-Mode}}{\text{Standard deviation}} \\ &= \frac{n - (n - 2)}{\sqrt{2n}} \\ &= \sqrt{\frac{2}{n}}.\end{aligned}$$

Since Pearson coefficient of skewness is greater than zero for $n \geq 1$, the χ^2 -distribution is positively skewed.

Further since skewness is inversely proportional to the square root of degrees of freedom, it rapidly tends to symmetry as the degrees of freedom increases.

2.7 Additive Property of χ^2 -distribution

The sum of independent χ^2 -variate is also a χ^2 -variate. More precisely, if X_i 's ($i = 1, 2, \dots, k$) are independent χ^2 -variates with n_i degrees of freedom respectively, then the sum $\sum_{i=1}^k X_i$ is also a χ^2 -variate with $\sum_{i=1}^k n_i$ degrees of freedom.

Proof: We have

$$M_{X_i}(t) = (1 - 2t)^{-\frac{n_i}{2}}; \quad i = 1, 2, \dots, k.$$

The moment generating function of the sum $\sum_{i=1}^k X_i$ is given by:

$$\begin{aligned}M_{\sum X_i}(t) &= M_{X_1}(t) \cdot M_{X_2}(t) \cdots M_{X_k}(t) \quad [\because X_i\text{'s are independent}] \\ &= (1 - 2t)^{-\frac{n_1}{2}} \cdot (1 - 2t)^{-\frac{n_2}{2}} \cdots (1 - 2t)^{-\frac{n_k}{2}} \\ &= (1 - 2t)^{-\frac{(n_1 + n_2 + \cdots + n_k)}{2}}\end{aligned}$$

which is the moment generating function of a χ^2 -variate with $(n_1 + n_2 + \cdots + n_k)$ degrees of freedom. Hence by uniqueness theorem of moment generating functions $\sum_{i=1}^k X_i$ is a χ^2 -variate with $\sum_{i=1}^k n_i$ degrees of freedom.

Assignment: Let X and Y are two independent χ^2 -variates with n_1 and n_2 degrees of freedom respectively, then prove the following:

- (i) $X + Y \sim \chi^2_{(n_1+n_2)}$,
- (ii) $\frac{X}{Y} \sim \beta_2\left(\frac{n_1}{2}, \frac{n_2}{2}\right)$, and
- (iii) $\frac{X}{X+Y} \sim \beta_1\left(\frac{n_1}{2}, \frac{n_2}{2}\right)$.

Note: The sum of two independent χ^2 -variates is also a χ^2 -variate and the ratio of two independent χ^2 -variates is a β -variate.

Example

If X is χ^2 -variate with n degrees of freedom, then prove that in case of large n , $\sqrt{2X} \sim N(\sqrt{2n}, 1)$.

Solution: Since $X \sim \chi^2_{(n)}$, we have the following:

$$E(X) = n \quad \text{and} \quad \text{Var}(X) = 2n.$$

and

$$\begin{aligned} Z &= \frac{X - E(X)}{\sigma_X} \\ &= \frac{X - n}{\sqrt{2n}} \sim N(0, 1), \text{ for large } n. \end{aligned} \tag{2.5}$$

Now consider the following

$$\begin{aligned} P\left(\frac{X - n}{\sqrt{2n}} \leq z\right) &= P\left(X \leq n + z\sqrt{2n}\right) \\ &= P\left[\sqrt{2X} \leq \left(2n + 2z\sqrt{2n}\right)^{\frac{1}{2}}\right] \\ &= P\left[\sqrt{2X} \leq \sqrt{2n} \left(1 + z\sqrt{\frac{2}{n}}\right)^{\frac{1}{2}}\right] \\ &= P\left[\sqrt{2X} \leq \sqrt{2n} \left(1 + \frac{z}{\sqrt{2n}} - \frac{z^2}{4n} + \dots\right)\right] \end{aligned}$$

$$\begin{aligned}
& \left[(1+x)^n = 1 + nx + n(n-1)\frac{x^2}{2!} + \dots \right] \\
& = P \left[\sqrt{2X} \leq \sqrt{2n} + z \right], \quad \text{for large } n \\
& = P \left[\sqrt{2X} - \sqrt{2n} \leq z \right], \quad \text{for large } n.
\end{aligned}$$

Since for large n , $\frac{X-n}{\sqrt{2n}} \sim N(0, 1)$, from (2.5) we conclude that

$$\sqrt{2X} - \sqrt{2n} \sim N(0, 1), \quad \text{for large } n$$

$$\therefore \sqrt{2X} \text{ is asymptotically } N(\sqrt{2n}, 1).$$

****Theorem**

Let $X_1, X_2, X_3, \dots, X_n$ be a random sample from a normal population with mean μ and variance σ^2 . Then

- (i) $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ and $\frac{(n-1)s^2}{\sigma^2} = \sum_{i=1}^n \left(\frac{X_i - \bar{X}}{\sigma} \right)^2$ are independently distributed.
- (ii) $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$.
- (iii) $\sum_{i=1}^n \left(\frac{X_i - \bar{X}}{\sigma} \right)^2 = \frac{(n-1)s^2}{\sigma^2} \sim \chi_{(n-1)}^2$.

Proof: (i). The joint probability differential of $X_1, X_2, \dots, X_n$ is given by

$$\begin{aligned}
dP(x_1, x_2, \dots, x_n) &= \left(\frac{1}{\sigma\sqrt{2\pi}} \right)^n e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2} \prod_{i=1}^n dx_i, \\
&\quad -\infty < (x_1, x_2, \dots, x_n) < \infty.
\end{aligned}$$

Let us transform to the variables Y_i , ($i = 1, 2, \dots, n$) by means of a linear orthogonal transformation $Y = AX$, where

$$\begin{aligned}
Y &= [y_1, y_2, \dots, y_n]', \\
X &= [x_1, x_2, \dots, x_n]', \\
A_{n \times n} &= [a_{ij}], \quad i, j = 1, 2, \dots, n.
\end{aligned}$$

Let us choose in particular:

$$\begin{aligned} a_{11} &= a_{12} = \cdots = a_{1n} = \frac{1}{\sqrt{n}} \\ \Rightarrow y_1 &= \frac{1}{\sqrt{n}} (x_1 + x_2 + \cdots + x_n) = \sqrt{n}\bar{x}. \end{aligned}$$

It can be easily seen that the above choice of $a_{11}, a_{12}, \dots, a_{1n}$ satisfies the condition of orthogonality, such that $\sum_{j=1}^n a_{ij}^2 = 1$. Since the transformation is orthogonal, we have the following

$$\begin{aligned} \sum_{i=1}^n y_i^2 &= \sum_{i=1}^n x_i^2 = \sum_{i=1}^n (x_i - \bar{x})^2 + n\bar{x}^2 \\ \Rightarrow \sum_{i=1}^n y_i^2 &= \sum_{i=1}^n (x_i - \bar{x})^2 + y_1^2 \\ \therefore \sum_{i=2}^n y_i^2 &= \sum_{i=1}^n (x_i - \bar{x})^2. \end{aligned}$$

Also

$$\begin{aligned} \sum_{i=1}^n (x_i - \mu)^2 &= \sum_{i=1}^n (x_i - \bar{x} + \bar{x} - \mu)^2 \\ &= \sum_{i=1}^n (x_i - \bar{x})^2 + 0 + \sum_{i=1}^n (\bar{x} - \mu)^2, \quad \left[\sum_{i=1}^n (x_i - \bar{x}) = 0 \right] \\ &= \sum_{i=2}^n y_i^2 + n(\bar{x} - \mu)^2. \end{aligned}$$

The Jacobian of the transformation $|J| = 1$. Thus the joint density function of $X_1, X_2, \dots, X_n$ transforms to:

$$\begin{aligned} dG(y_1, y_2, \dots, y_n) &= \left(\frac{1}{\sigma\sqrt{2\pi}} \right)^n e^{-\frac{1}{2\sigma^2} \{ \sum_{i=2}^n y_i^2 + n(\bar{x} - \mu)^2 \}} \cdot |J| \cdot \prod_{i=1}^n dy_i, \\ &= \left[\frac{1}{\frac{\sigma}{\sqrt{n}}\sqrt{2\pi}} e^{-\frac{n}{2\sigma^2} (\bar{x} - \mu)^2} d\bar{x} \right] \quad ; [dy_1 = \sqrt{n}d\bar{x}] \\ &\quad \times \left(\frac{1}{\sigma\sqrt{2\pi}} \right)^{n-1} e^{-\frac{1}{2\sigma^2} \sum_{i=2}^n y_i^2} \prod_{i=2}^n dy_i. \end{aligned} \quad (2.6)$$

Thus $\bar{X}$ and $\sum_{i=2}^n Y_i^2 = \sum_{i=1}^n (X_i - \bar{X})^2 = (n-1)s^2$ are independently distributed.

(ii). From (2.6), it is very much clear that

$$dP(\bar{x}) = \frac{1}{\frac{\sigma}{\sqrt{n}}\sqrt{2\pi}} e^{-\frac{n}{2\sigma^2}(\bar{x}-\mu)^2} d\bar{x}, \quad -\infty < \bar{x} < \infty,$$

where $E[\bar{X}] = \mu$ and $Var[\bar{X}] = \frac{\sigma^2}{n}$. Hence $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$.

(iii). From (2.6), it is also clear that Y_i , ($i = 2, \dots, n$) are independent $N(0, \sigma^2)$. Hence, we can write as

$$\sum_{i=2}^n \frac{y_i^2}{\sigma^2} = \sum_{i=1}^n \left(\frac{X_i - \bar{X}}{\sigma} \right)^2 = \frac{(n-1)s^2}{\sigma^2},$$

being the sum of squares of $(n-1)$ independent standard normal variates is distributed as χ^2 -variate with $(n-1)$ degrees of freedom.

2.8 Applications of χ^2 -distribution

In Statistics, χ^2 -distribution has a large number of applications, some of which are given as:

- (i) test the goodness of fit,
 - (ii) test the independence of attributes,
 - (iii) test the homogeneity of independent estimates of the population variance and population correlation coefficient,
 - (iv) in confidence interval estimation for a population standard deviation of a normal distribution from a sample standard deviation,
- and so on.